

EEG Data Analytics Dashboard

Muse Headband Brainwave Analysis Project

1. Project Overview

This project focuses on EEG (Electroencephalography) brainwave data analysis using the Muse Headband device. It demonstrates a complete end-to-end analytics workflow encompassing data collection, preprocessing, transformation, exploratory data analysis (EDA), ETL processes, and dashboard visualization using Power BI.

2. Project Objectives

- Collect EEG brainwave data using Muse Headband
- Clean and preprocess raw EEG datasets for high-quality reporting
- Transform datasets into an analytical structure suitable for deep-dive analysis
- Perform Exploratory Data Analysis (EDA) to understand signal behavior
- Develop interactive Power BI dashboards for real-time monitoring
- Visualize EEG activity, signal behavior, and generate actionable analytical insights

3. Dataset Information

The dataset contains electrode signal readings captured from multiple brainwave channels across different regions of the brain.

EEG Channel	Description
AF7	Frontal brain activity signal
AF8	Frontal brain activity signal
TP9	Temporal region signal
TP10	Temporal region signal

4. Data Preprocessing & ETL

Data Preprocessing Steps:

- Handled missing and null values to ensure data integrity
- Removed unnecessary columns to streamline the analysis
- Structured and formatted records for optimal visualization in Power BI
- Prepared transformed datasets using Excel for reporting quality

ETL Workflow:

Stage	Activities
Extract	Collected EEG data from Muse Headband
Transform	Cleaned and transformed EEG datasets in Excel
Load	Loaded transformed data into Power BI for visualization

5. Exploratory Data Analysis (EDA)

EDA was performed to evaluate electrode performance and identify trends through:

- Signal trend analysis and Time-series EEG visualization
- Electrode comparison and signal activity observation
- Pattern and variability analysis across different brainwave channels

6. Dashboard & Technology

Dashboard Features:

- EEG signal monitoring and Brainwave heatmap visualization
- Electrode channel analysis with interactive slicers and filters
- KPI cards for core metrics and time-series signal visualizations
- Professional layout optimized for analytical reporting

Tools & Technologies:

Tool	Purpose
Muse Headband	EEG Data Collection
Excel	Data Cleaning & Transformation
Power BI	Dashboard Development & Visualization

7. Conclusion

This project successfully demonstrates a complete EEG analytics workflow. By combining robust preprocessing with dynamic Power BI visualizations, the final dashboard provides professional insights into brainwave activity, highlighting practical skills in data engineering and analytical reporting.